Two Optimizer Sketches for QiOpt4QNet

Hamiltonian/Metropolis Annealing vs. Tensor-Network Compression
(inspired by arXiv:2605.27425)

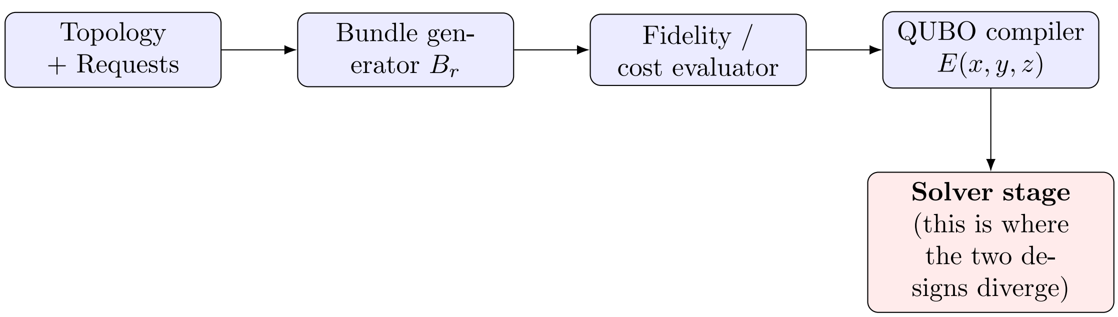
QIntern 2026 – QiOpt4QNet

July 2026

This note sketches two concrete optimizer designs for the QiOpt4QNet bundle-selection QUBO (Sec. 5.3–5.4 of the project proposal), following the two solver families used in arXiv:2605.27425 for QKD routing: a **Hamiltonian / Metropolis annealer** and a **tensor-network (boundary-MPS) compressor**. Both share the same front end and differ only in how the low-energy bundle-selection configuration is searched for.

1 Shared Front End

Both optimizers consume the same problem instance: a topology, a request batch, generated candidate bundles B_r, and the compiled QUBO energy $E(x, y, z)$ from Sec. 5.4 of the proposal. Only the "solve" stage differs.



2 Optimizer A – Hamiltonian / Metropolis Annealer

2.1 Idea

Treat a bundle-selection configuration $x = \{x_{(r,b)}\}$ as a physical microstate of the network. Its energy is exactly the QUBO objective:

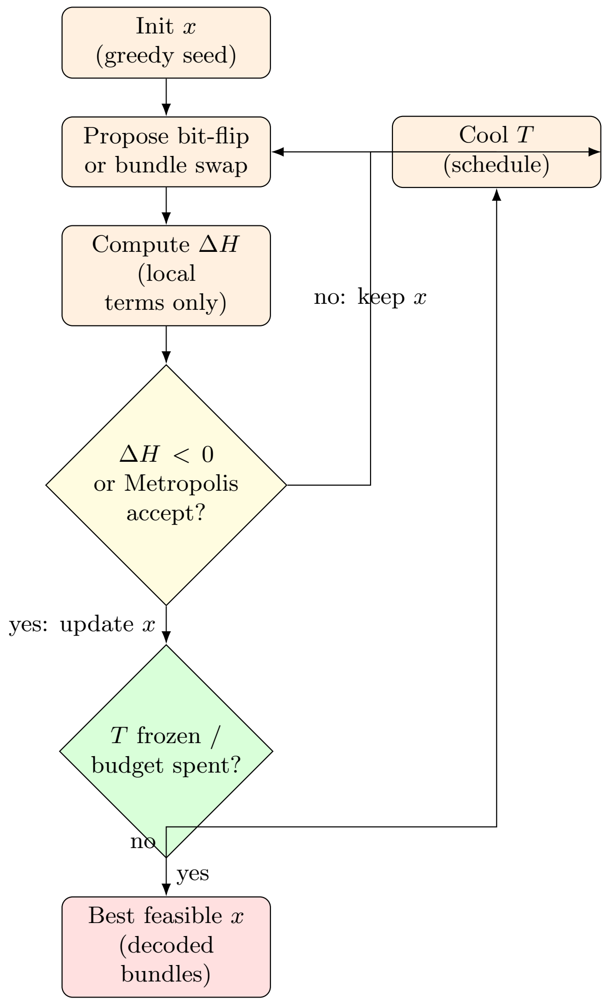
$$H(x) \;=\; E(x, y, z) \;=\; -\sum_{r,b} u_{(r,b)} x_{(r,b)} + A \sum_{r} (\cdots) + B \sum_{e} (\cdots)^2 + D \sum_{v} (\cdots)^2.$$

Good routing/allocation decisions are low-energy states. A Metropolis Monte-Carlo sampler explores configuration space with local bit-flip moves, exactly as in arXiv:2605.27425's Metropolis annealer for QKD routing energy.

2.2 Loop

1. Initialize x (e.g. all-reject, or greedy utility-density seed).
2. Propose a local move: flip one $x_{(r,b)}$, or swap the active bundle for one request r (single-choice constraint aware move).
3. Compute ΔH from the local energy change only (cheap, since H is a sum of local + pairwise-capacity terms).

4. Accept if $\Delta H < 0$; otherwise accept with probability $e^{-\Delta H/T}$.
5. Cool T on a schedule (geometric or adaptive); repeat until frozen.
6. Return the lowest-energy feasible configuration seen.



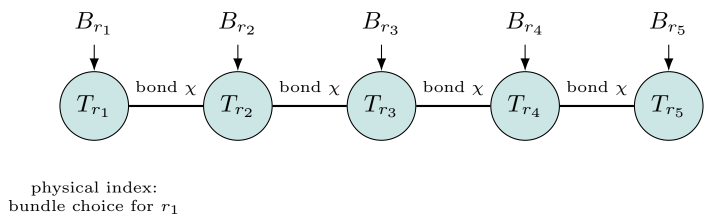
2.3 Role in QiOpt4QNet

This *is* our simulated-annealing / tabu baseline from Sec. 5.5, made explicit as a physical annealing loop. Its main value: a transparent, easily instrumented baseline against OpenJij/neal, and a natural place to plug in the paper's exact Metropolis acceptance rule for a controlled, apples-to-apples comparison.

3 Optimizer B – Tensor-Network (Boundary-MPS) Compressor

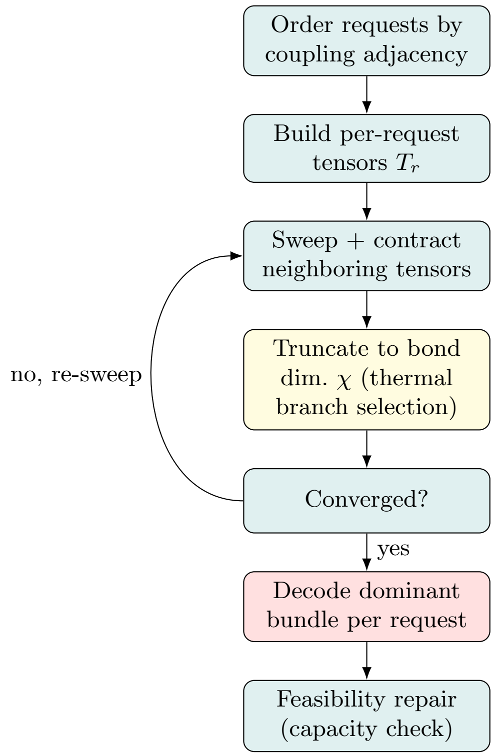
3.1 Idea

Instead of sampling configurations one at a time, represent the *joint distribution* over all requests' bundle choices as a **Matrix Product State (MPS)** – a chain of small tensors, one per request r, each with a physical index ranging over $B_r \cup \{\text{reject}\}$ and virtual "bond" indices carrying correlations to neighboring requests (shared edges/memories). Contracting and truncating this chain approximates the low-energy (high-utility, feasible) sector directly, without exhaustive sampling – the same boundary-MPS compression arXiv:2605.27425 uses for QKD route enumeration.



3.2 Loop

1. Order requests $r_1, \ldots, r_{|R|}$ so that requests sharing edges / memories (strong QUBO couplings) are adjacent in the chain.
2. Build one rank-3 tensor T_r per request from local utility $u_{(r,b)}$ and its coupling terms to neighbors.
3. Sweep left-to-right (and back), contracting neighboring tensors and **truncating** to bond dimension χ, keeping only the highest-weight (lowest-energy) branches – the "thermal branch selection" step.
4. After convergence, read off the dominant bundle choice per request by contracting the MPS with a one-hot probe at each site.
5. Feasibility repair pass: fix any residual capacity violations left by the compression (rare, since strong couplings are captured by χ).



3.3 Role in QiOpt4QNet

This is the *new* idea contributed by the paper: instead of only k-shortest-path pruning of candidate bundles, use bond-dimension truncation to compress correlated request groups (those competing for the same edge or memory) before/instead of full QUBO sampling. Proposed as an ablation (Sec. 7.3): compare *annealer-only*, *tensor-compression-only*, and *tensor-compression* $\rightarrow$ *annealer refine* pipelines on the same benchmark instances.

4 Side-by-Side

	A: Hamiltonian / Metropolis	**B: Tensor-network (boundary-MPS)**
Search style	Sequential local sampling	Global compressed contraction
Cost driver	# sweeps $\times$ # variables	bond dimension χ
Best suited to	Small/medium instances, fast baselines	Instances with strong local coupling structure (dense contention)
Failure mode	Slow mixing at low T	Truncation loses rare but important branches
QiOpt4QNet role	Baseline solver (Sec. 5.5)	New scalability ablation (Sec. 7.3)

Suggested experiment: run both optimizers on identical chain/grid topologies and request loads (Sec. 7.1), report served-request ratio, aggregate utility, and wall-clock time (Sec. 7.2), and use the results to decide whether tensor-network compression is worth the added implementation complexity for QiOpt4QNet's benchmark suite.